

Proximal Drift-Plus-Penalty

Time-Varying Constraints: $h_t = h + \psi_t$

September 30, 2026

Primal problem (time-varying):

$$\min_{x \in X} f(x) \quad \text{s.t.} \quad g(x) \leq 0, \quad \limsup_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T h_t(x_t) \leq 0$$

Penalized objective:

$$P(x) = f(x) + \beta [g(x)]_+$$

Algorithm (per step t , $q_t := Q_t/V$, observing h_t):

$$x_{t+1} = \arg \min_{x \in X} P(x) + q_t (h_t(x) + \gamma \|x - x_t\|^2) + \frac{1}{2\eta} \|x - x_t\|^2$$

$$Q_{t+1} = \max(Q_t + h_t(x_{t+1}), 0)$$

Domain and Function Definitions

$$X = \{x \in \mathbb{R}^d : \|x\| \leq R_X\} \quad D := \text{diam}(X) = 2R_X$$

$$f(x) = \frac{1}{2} w_{\text{quad}}^f \|x\|^2 - c_f^\top x + \sum_{j=1}^6 \alpha_j^f \cos(w_j^f \cdot x + b_j^f)$$

$$g(x) = \|x\|^2 - r_g^2$$

$$h_t(x) = \underbrace{\frac{1}{2} w_{\text{quad}}^h \|x\|^2 + \sum_{j=1}^3 \alpha_j^h \cos(w_j^h \cdot x + b_j^h) + \text{offset}_h}_{h(x), \text{ fixed}} + \underbrace{\psi_t}_{\text{drifts each round}}$$

- g : exactly convex ball: kept this way so $\kappa_g = 2\beta r_g$ (Assumption 6) holds
- ψ_t : additive scalar drift, constant in x : does not change h 's shape, only its offset

Synthetic Instance: Exact Parameters

$d = 10$, $R_X = 1$, $r_g = 0.75$, $\xi = 0.4$, $D = 2$, $\text{seed} = 0$

$w_{\text{quad}}^f = 0.2$, $\|c_f\| = 0.5$ $w_{\text{quad}}^h = 2.0$, $\text{offset}_h = -0.400722$

f : 6 cosine terms

j	1	2	3	4	5	6
α_j^f	0.20	0.20	0.20	0.20	0.20	0.20
$\ w_j^f\ $	1.00	1.24	1.48	1.72	1.96	2.20
b_j^f	4.00	1.70	0.26	0.10	5.11	5.74

h : 3 cosine terms

j	1	2	3
α_j^h	0.025	0.025	0.025
$\ w_j^h\ $	1.00	1.25	1.50
b_j^h	5.96	2.89	4.76

(direction vectors $w_j^f/\|w_j^f\|$, $w_j^h/\|w_j^h\|$, and $c_f/\|c_f\|$: random unit vectors, seed=0 – omitted here, 10 components each)

$$\Rightarrow \rho_f = 3.274, \quad \rho_h = 0.120, \quad L_f = 2.620, \quad L_h = 2.094, \quad H = 1.476, \\ G_g = 0.438$$

Hyperparameters

Algorithm:

β	γ	η	V	T
679.0	0.065	0.2	20.0	4000

Derived windows (must contain the chosen γ, η):

$$\gamma \in \left(\frac{\rho_h}{2}, \frac{\xi}{D^2} \right) = (0.060, 0.100) \qquad \eta \in \left(0, \frac{1}{\rho_f} \right) = (0, 0.305)$$

Drift ψ_t :

$\psi_{\max}$ (valid)	$\psi_{\max}$ (Slater-violating)	drift seed
0.08	0.6	11

Sweep grids:

- $V \in \{0.1, 0.25, 0.5, 1, 2, 5, 10, 20, 50, 100\}$
- $\beta \in \{0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 1, 5, 20, 50, 100, 300, 679, 800\}$

The Drift Mechanism

ψ_t is generated *once*, before the run starts – the algorithm only ever sees ψ_t at round t , never the future.

Slow drift (small upward bias + noise, clipped):

$$\psi_{t+1} = \text{clip}(\psi_t + 0.0006 + \mathcal{N}(0, 0.012^2), 0, \psi_{\max})$$

Bursts (mostly flat, occasional jumps that decay back down):

$$\text{level}_{t+1} = 0.997 \cdot \text{level}_t + \mathbb{I}[\text{burst at } t] \cdot \psi_{\max} \cdot \text{Unif}(0.5, 1.0)$$

$$\psi_t = \text{clip}(\text{level}_t + \mathcal{N}(0, 0.004^2), 0, \psi_{\max})$$

Functional variation (what the theory's V_T term measures):

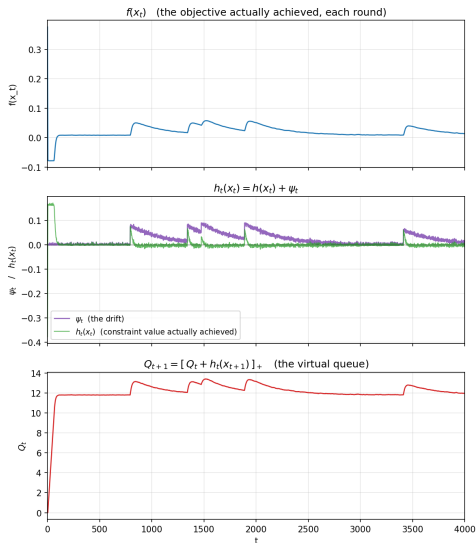
$$\Delta_t = |\psi_{t+1} - \psi_t| \qquad V_T = \sum_{t=1}^{T-1} \Delta_t$$

Why ψ_t alone determines Δ_t

$h_{t+1}(x) - h_t(x) = \psi_{t+1} - \psi_t$ for every x – the $h(x)$ part cancels exactly, so the functional variation reduces to a 1-D quantity.

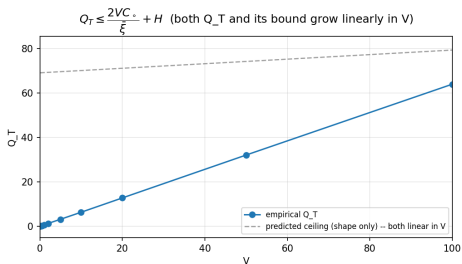
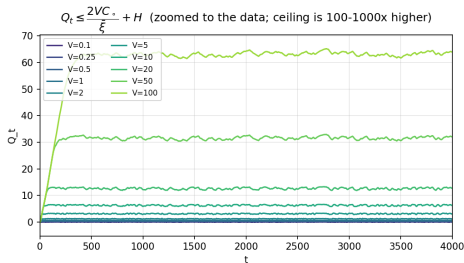
One Trajectory (Burst Drift)

One trajectory, burst drift: f dips while the queue is empty, then pays it back once a burst raises h_t

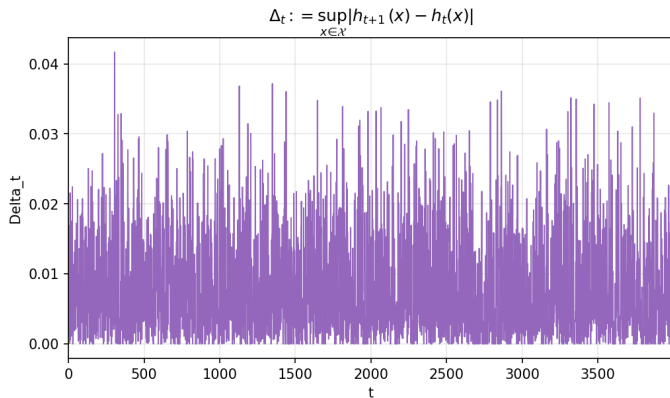


Lemma 2: Pointwise Queue Bound

Lemma 2: pointwise queue bound, time-varying h_t

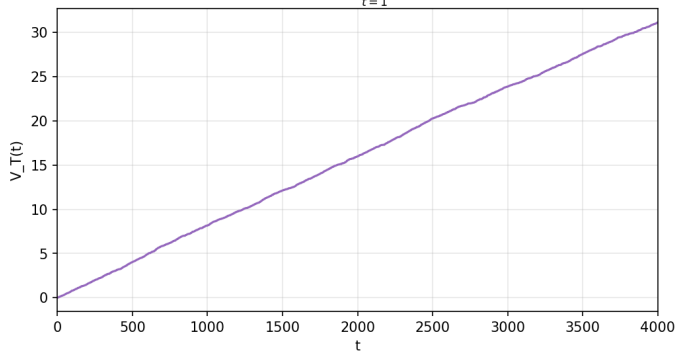


Definition 1: One-Step Functional Variation

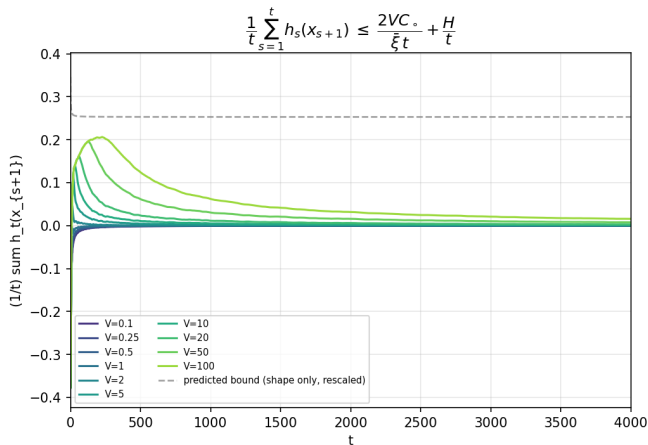


Definition 1: Cumulative Functional Variation

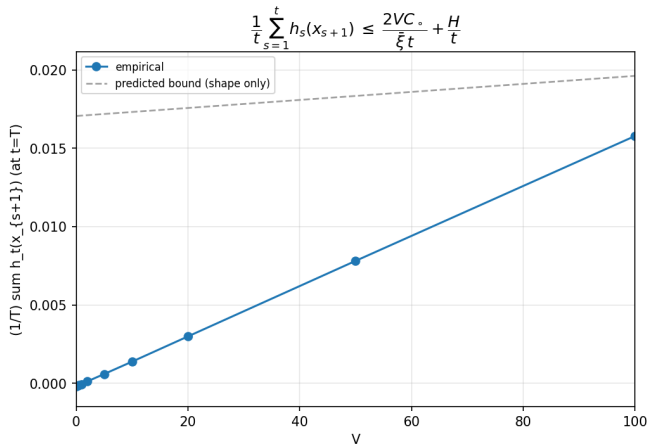
$$V_T := \sum_{t=1}^{T-1} \Delta_t$$



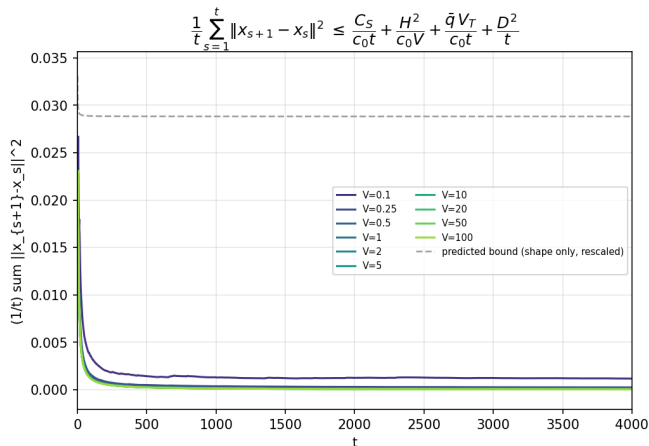
Theorem 1: Time-Average Feasibility, vs t



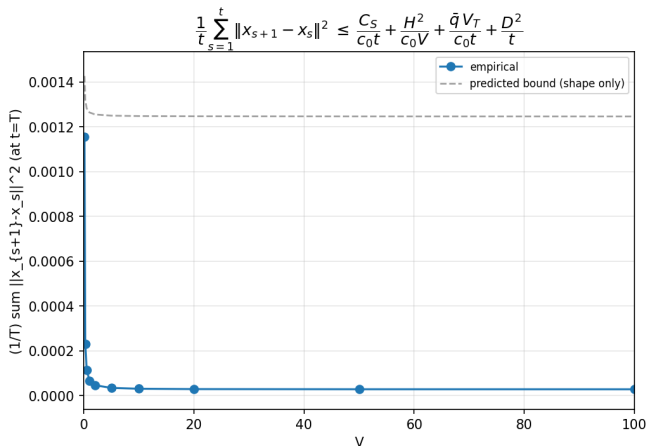
Theorem 1: Time-Average Feasibility, vs V



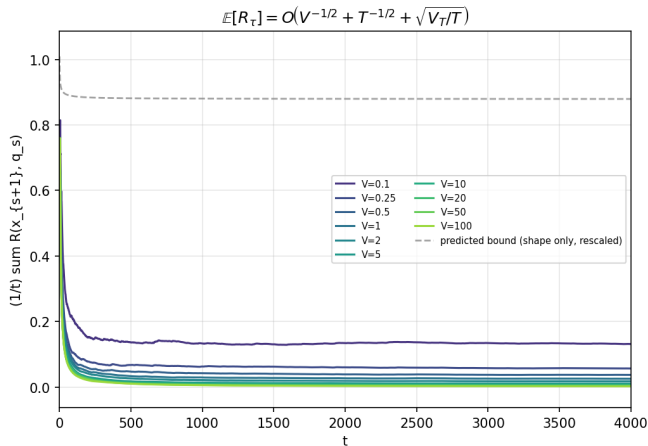
Theorem 2 (Movement), vs t



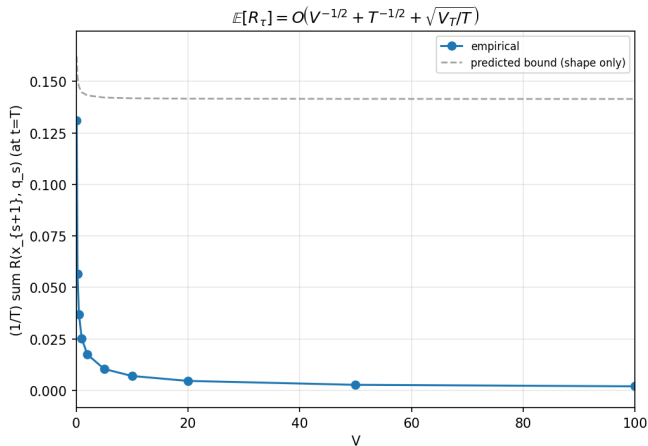
Theorem 2 (Movement), vs V



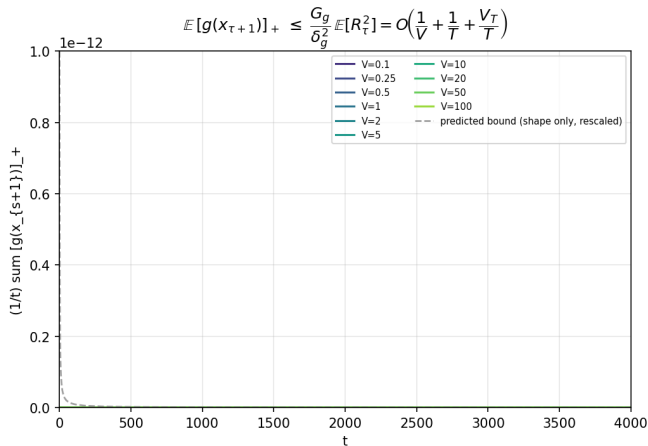
Theorem 2 (Near-Stationarity Residual), vs t



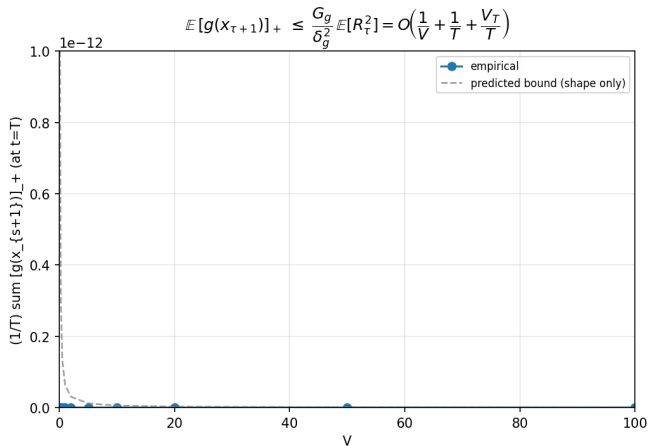
Theorem 2 (Near-Stationarity Residual), vs V



Theorem 3 (Nearly-feasible stationarity for the ordinary constraint), vs t

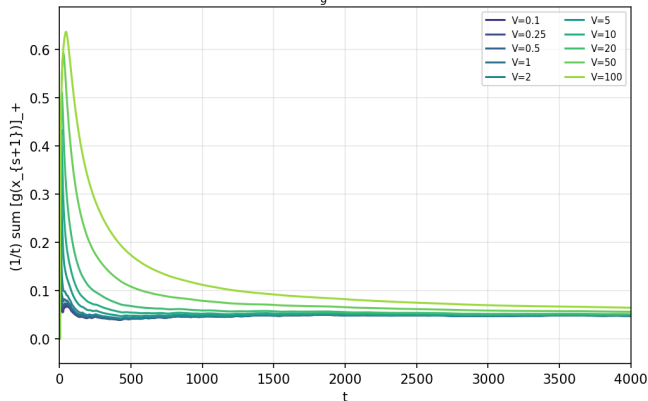


Theorem 3 (Nearly-feasible stationarity for the ordinary constraint), vs V

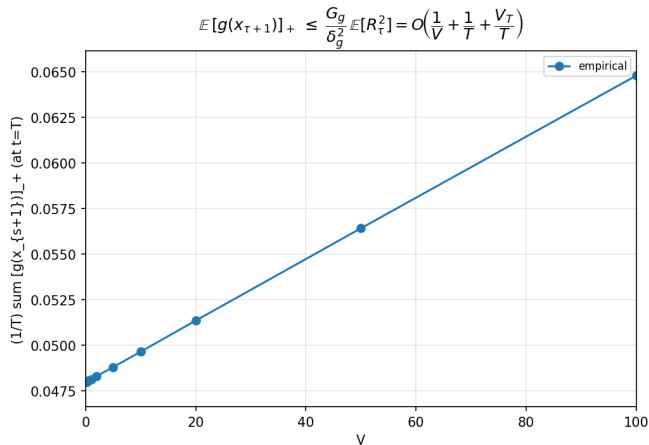


Theorem 3 (Dangerous Instance, $\beta = 0.4$), vs t

$$\mathbb{E}[g(x_{\tau+1})]_+ \leq \frac{G_g}{\delta_g^2} \mathbb{E}[R_\tau^2] = \mathcal{O}\left(\frac{1}{V} + \frac{1}{T} + \frac{V_T}{T}\right)$$



Theorem 3 (Dangerous Instance, $\beta = 0.4$), vs V

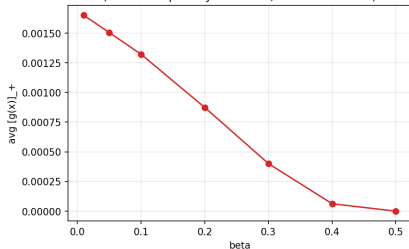


Bigger V weakens the queue's restraint on h_t (which also pulls inward here) while β 's restraint stays fixed – so violation *grows* with V , opposite the theorem's own $O(1/V)$ rate, which assumed $\delta_g > 0$.

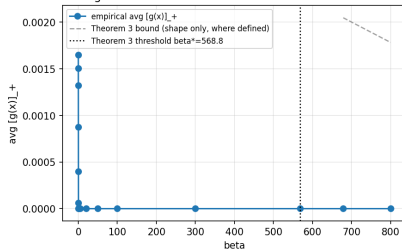
Theorem 3: β Sweep (Original Instance)

Theorem 3: violating beta -- the real threshold vs. the proved one

Small beta: the ordinary constraint is actually violated
(real exact-penalty threshold, not Theorem 3's)



Full beta range: Theorem 3's sufficient threshold is far more conservative



Real threshold (≈ 0.5) vs. Theorem 3's own threshold ($\beta^* \approx 569$) – about $1100\times$ apart, from stacking a worst-case $\bar{q}$, Markov's inequality, and a worst-case regularity constant

κ_g .